Data Integrity Report

This report covers the review of task1_combined_data.csv (417 checkout records, 18 columns) before it was used for any further analysis. Four data quality problems were found and corrected. The cleaned file is saved as task2_cleaned_data.csv (404 records).

1. Missing Values

What was found

Eight columns had missing values, but they fell into two very different groups. Four columns were missing because the underlying information genuinely doesn't exist yet or was never recorded. Four more were missing only because the checkout belonged to a member_id that doesn't exist in the members table (Problem 4) — once those rows were removed, this second group of gaps disappeared on its own.

Where and how big

Column	Missing (of 417)	Reason
return_date	91	Book not yet returned, or a Reading Kickoff checkout (no return date is ever recorded for those)
grade	41	5 rows tied to Problem 4; the rest are registered members with no grade on file
join_date	11	Registered members with no join date on file
publication_year	35	3 titles in the book catalog have no publication year recorded
first_name / last_name / neighborhood / membership_status	5 each	All 5 rows tied to Problem 4 (invalid member_id)

What was done, and why

return_date, grade, join_date and publication_year were left blank. None of these gaps can be filled in reliably: a blank return_date means the book is still out, and there is no honest way to guess a member's grade, join date, or a book's publication year. Filling them in would mean inventing data rather than reporting what's actually known. The missing name/neighborhood/status values were resolved as a side effect of removing the 5 invalid member_id rows in Problem 4, rather than being treated as a separate fix.

2. Duplicate Records

What was found

8 checkout_id values (9052, 9180, 9193, 9194, 9246, 9280, 9296, 9334) each appeared twice, with every column identical — same member_id, book_id, checkout_date and return_date. These are the same checkout event recorded twice, not two separate borrowings.

This is different from 188 other rows in the dataset where the same member borrowed the same book more than once, but on different dates with different checkout_id values (for example, member 1008 borrowing book 501 three separate times). Those are real, distinct checkouts and were left untouched.

Where and how big

Column affected: the checkout record as a whole (all 18 columns matched). 8 duplicate rows were found and removed.

What was done, and why

One copy of each of the 8 repeated rows was removed, keeping a single record per real checkout event. They were identified by matching on every column at once, which is why the 188 legitimate repeat-borrow rows (same member and book, different date) were correctly left in the dataset instead of being mistaken for duplicates.

3. Inconsistent Text Values

What was found

Two columns had the same real-world value written in more than one way: neighborhood and membership_status.

Column	Forms found	Rows affected
neighborhood	"NASR CITY" vs "Nasr City"; "HELIOPOLIS" vs "Heliopolis"; "Maadi " (trailing space) vs "Maadi"; "zamalek" vs "Zamalek"	31
membership_status	"active" vs "Active"; "inactive" vs "Inactive"	78

One value was deliberately left alone: the genre column contains both "Science" and "Science Fiction". These look similar but describe two different genres, so they were kept separate rather than merged into one.

What was done, and why

Both columns were trimmed of extra spaces and converted to a single consistent capitalization (e.g. "Heliopolis", "Active"). This groups together values that are the same thing written differently, without touching values that are actually different (like the two genres above).

4. Invalid member_id

What was found

5 checkout records reference a member_id (1104, 1150 and 1201) that does not exist in the members table. All 5 come from the Reading Kickoff page, where students could check out a book without presenting a library card — so these checkouts were never tied to a registered member in the first place.

Where and how big

Column affected: member_id. 5 checkout records were affected, referencing 3 distinct invalid IDs.

What was done, and why

These 5 records were removed from the checkout dataset. The members table itself was not changed in any way — no placeholder members were added. This was chosen over keeping the rows because a checkout that can't be tied to a real member profile carries no usable member information (name, grade, neighborhood and status would all be blank) and would only add noise to any member-level analysis performed later. Removing them affects a small, clearly identified slice of the data (5 of 417 rows, about 1%) and does not affect the integrity of the members table.

Summary

Problem	Rows affected	Action
Missing values (4 genuine-gap columns)	return_date 91, grade 41, join_date 11, publication_year 35	Left blank
Duplicate records	8	Removed
Inconsistent text	31 (neighborhood) + 78 (membership_status)	Standardized
Invalid member_id	5	Removed

Starting rows: 417 → Final rows in task2_cleaned_data.csv: 404